

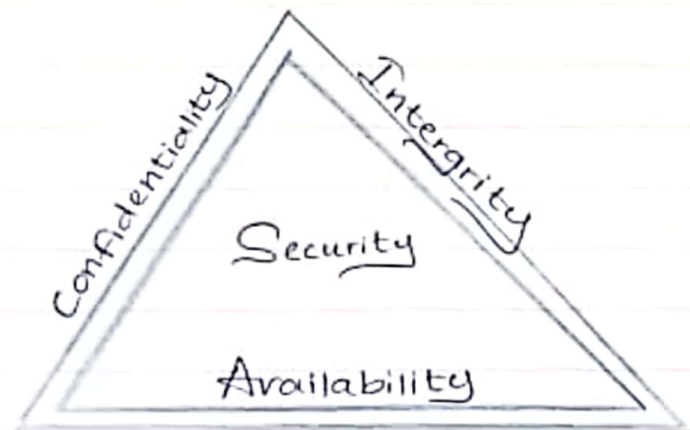
Answer the Questions.

01. What is Computer Security?
02. Explain the important assets used in computer systems
03. Briefly explain
 - Vulnerabilities
 - Threat
 - Attack
 - Control
04. Explain CIA Triangle?
05. Explain important characteristics of Authentication & non repudation?
06. Explain Security policy with main three contents?
07. What are the 03 types of threats explain with suitable examples?
08. What are the important 03 goals of Security? Explain with examples?

Answers

- 01 Computer Security is a protection of Unauthorized access that asset of Computer System as Hardware, Software, Data.
- 02 The important asset used in Computer Systems are as follows hardware, Software and Data. Their hardware consists of Computer devices and networks. While Software has Operating System, Utilities, translation and application. In addition the data provide Email, documents, photos etc. And Hardware and Software are easily replaceable (off the shelf) while Data is Unique and irreplaceable.
- 03 Vulnerability
← Weakness in an information system that cyber criminals can exploit and gain unauthorized access to computer system.
- Threat
← An potential violation of security, which exist when there is a circumstances that could breach security and cause harm.
- Attack
← Assault on system security that derives from an intelligent that within a deliberate attempt.
- Control
← Is an action, device, procedure or techniques that remove or reduce of vulnerability.

- 04 CIA Triad acronym for Confidentiality, Integrity and Availability that comprising as three core principles



Confidentiality is biased on keeping sensitive information which computer related assets are accessed only by authorized parties meanwhile Integrity ensures the reliability and accuracy of data that assets can modified only by authorized parties. Further availability depicts that access are accessible to authorized parties at appropriate time.

05. Authentication

← enables Organization to keep their network secure by permitting only authenticated users to gain access (permission) to the protected resources. This show up something reliable and valid.

Non repudiation.

← Probably digital signatures are common technology used to achieve non repudiation. This uniquely link a sender and a receiver. It means sending message

from one place to another. This assurance that someone cannot deny something.

06. Security policies are determined as follow.

- Subject (who)
- Mode of access (How)
- Object (what).

07. Natural Cause.

- This may be happens through eventually as fire, Power Failure

Human Cause.

- This may occur through human error.

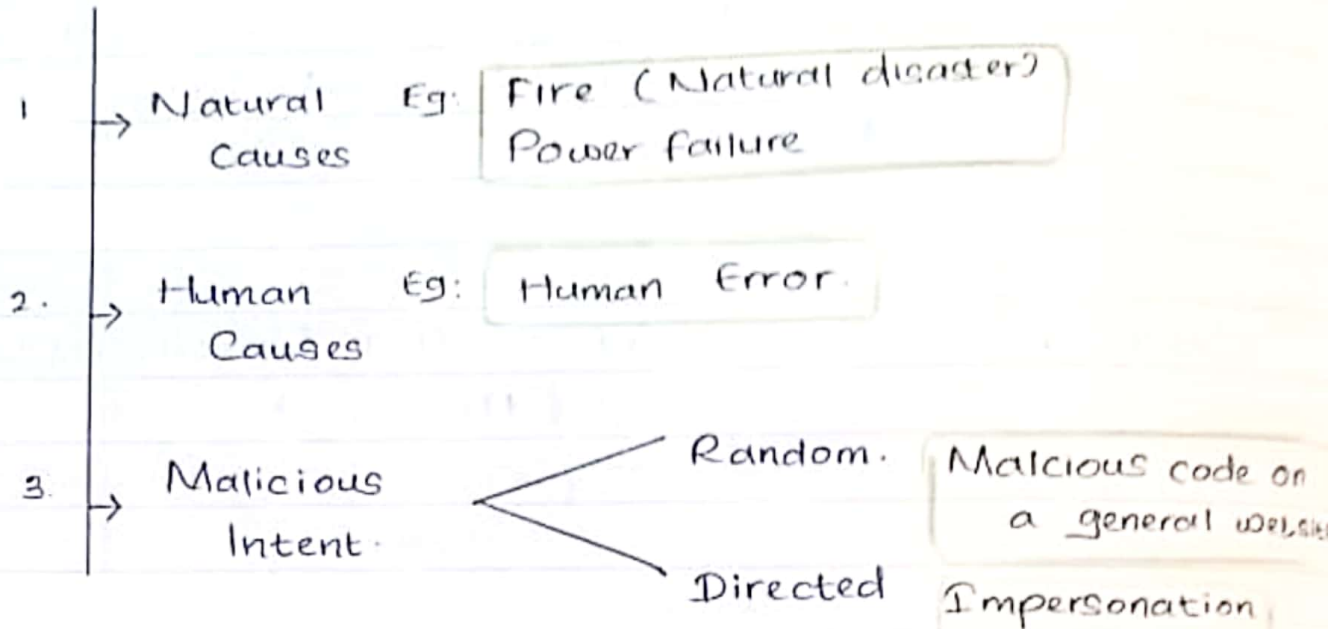
Malicious Attack.

- This happens through impersonation.

08. The important 3 goals of security are prevention, detection and recovery. Prevention is prevented through attackers through violating security as an example antivirus guard can be initially scanned and next it can be identify the issues and go through detection security policy to detect errors then it will send out to recovery that stop attack and recover the damages.

1. Computer Security → The protection of computer Systems and information from harm, theft and unauthorized use
The protection of the assets of a computer
 - Hardware
 - Software
 - Data
2. Value of assets → Off the shelf (Easily replaceable)
(Hardware & Software)
→ Unique (Irreplaceable)
(Data)
3. Vulnerability → Weakness in Security System that might exploit to cause loss or harm.
4. Threat → Potential cause loss or harm.
5. Attack → Assault on System Security
5. Important of Computer Security?
 - ▲ Prevention of data theft
 - ▲ Malicious attacks
(virus, worms, ransomware, Trojan horse, Phishing, Spoofing, Adware, Spyware, Keylogger, Bots)
 - ▲ Allow programs to run smoother & faster
7. Authentication → Process of verifying the identity of the sender or the origin of data. (Technical concept)
8. Non repudiation → legal concept Anyone can validate the authenticity of a message as well as source of the message It is based on digital signature (Public key cryptography)

Types of threats



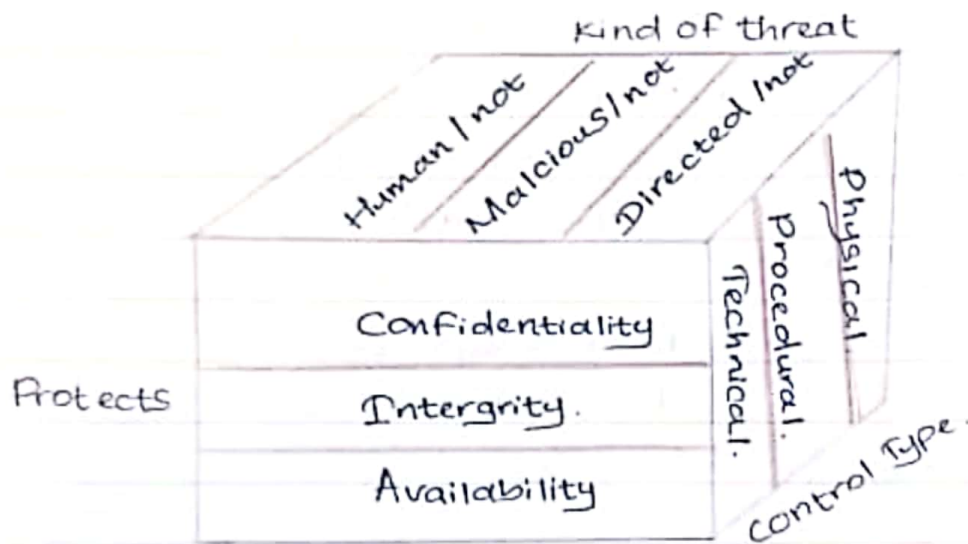
Types of Attackers

- ▲ Terrorist
- ▲ Hackers
- ▲ Individual
- ▲ Organized crime members.
- ▲ Criminal for hire.

Access Control

- ▲ Policy
- ▲ Subject → who.
- ▲ Access of mode → How.
- ▲ Object → what.

Controls / Countermeasures



Security goals

- ▲ Confidentiality.
- ▲ Integrity.
- ▲ Availability.

Vs.

Goals of Security

- ▲ Prevention.
- ▲ Detection.
- ▲ Recovery.

Computer Security challenges

- ▲ Not Simple.
- ▲ Cyber attacks
- ▲ Battles of wits between admin / attacker.
- ▲ Must consider potential attacks

OSI LAYERS "All people should try new dominos pizza"

7. Application layer (Data)
6. Presentation layer (Data)
5. Session layer (Data)
4. Transport layer (Segment) TCP/UDP
3. Network layer (Packet)
2. Data link layer (Frame)
1. Physical layer (Bits)

- OPEN SYSTEM INTERCONNECTION
- Developed by (ISO)
- "International Organization of Standardization"
- Bottom up approach
- Facilitate communication between two systems

TCP / IP LAYERS.

Application layer.
Transport layer.
Internet layer.
Network Access layer.

Access Control.

Policy

Subject.

→ Who

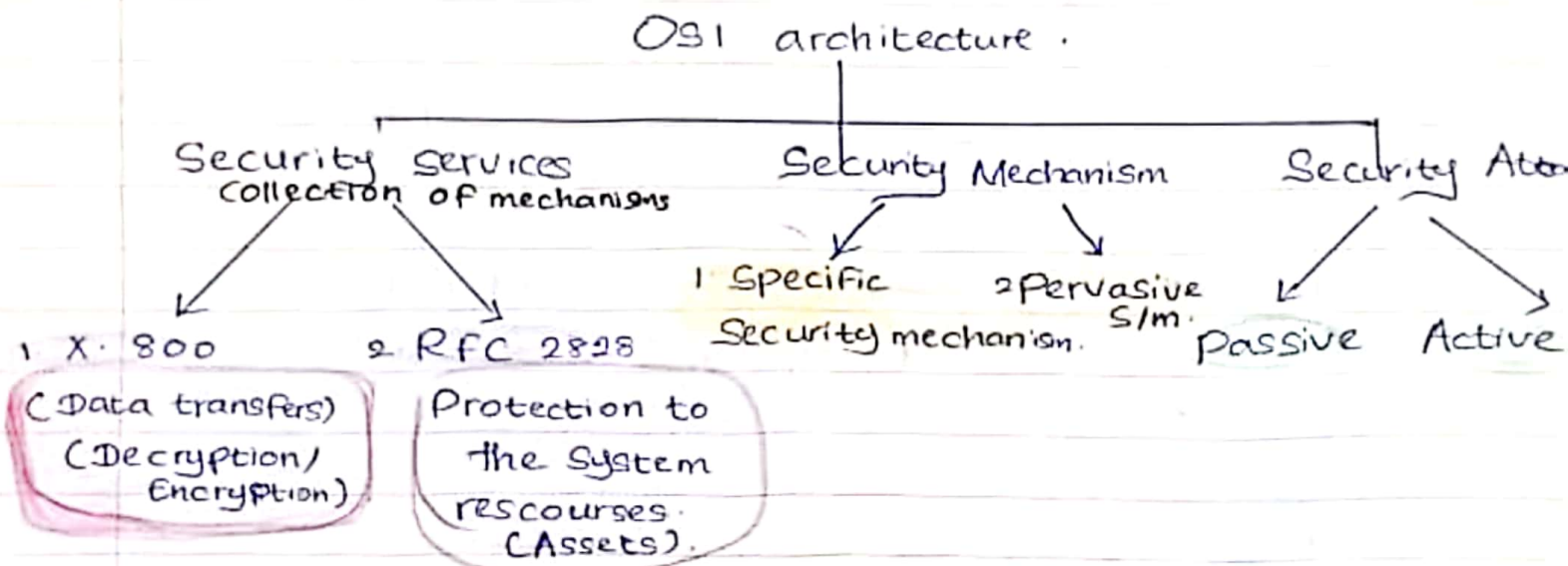
Mode of Access

→ How

Object

→ What.

OSI Security Architecture.



▲ Cryptography

- Apply the security.
- Encryption / Decryption

▲ Crypto Analysis.

Break the mechanism and read the context of document.

X.800

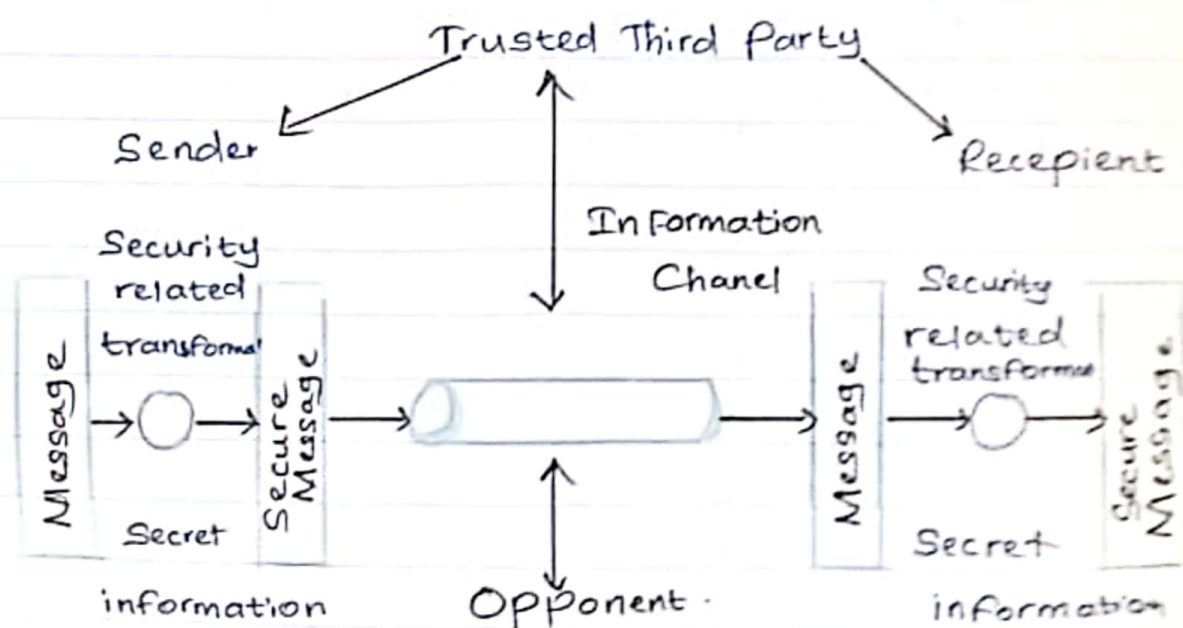
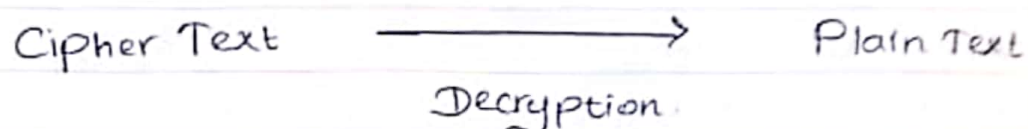
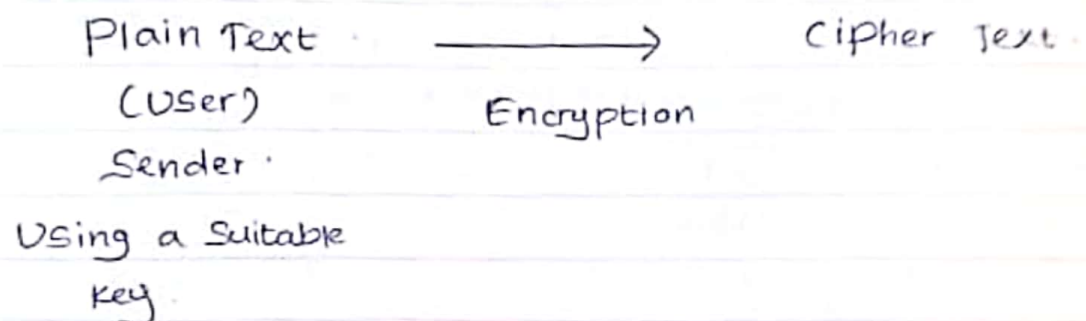
1. Authentication.
2. Control Access.
3. Data Confidentiality.
4. Data integrity.
5. Non repudation.

Inclusive Methods (detection)

Methods that use to identify the
Eg: Honey-pot Method.

Model for network security.

Opponent (Hacker)



Service, Mechanisms, Attacks

▮ Need systematic way to define requirements

▮ Consider the three important factors of information

Security

1. Security Attack
2. Security Mechanism.
3. Security Service.

Security Service.

* Security Service is something that enhance the Security of the data processing system & the information transfers of an organization.

* Intend to counter security attacks

* Make use of one or more security mechanisms to provide the service.

* Physical document use different methods to apply the security. Such as,

▮ Signature

▮ Dates

▮ need protection from disclosure

▮ Destructions.

▮ Unauthorized access or observed.

Security Attack.

* Any action that concerned the Security of information on while an Organization.

* Information Security is about how to prevent attacks. Or Failing that, to detect attacks on information based systems.

* Have a wide range of attacks.

* Can focus of general type of attacks.

(Regularly threat & attack meant same)

Security Mechanism.

* Mechanism that is designed to detect, prevent and recover from Security attacks

* No single mechanism that will support all functions required.

* However One specific element causes to many of the Security mechanisms.

Eg : Cryptographic techniques.

OSI Security Architecture.

▲ Define a Systematic way of defining and providing Security requirements.

Security Services

1. X.800

▲ Services provide by a protocol layer. of Communicating Open System. Which ensure acceptable Security of the System. or of data transfers

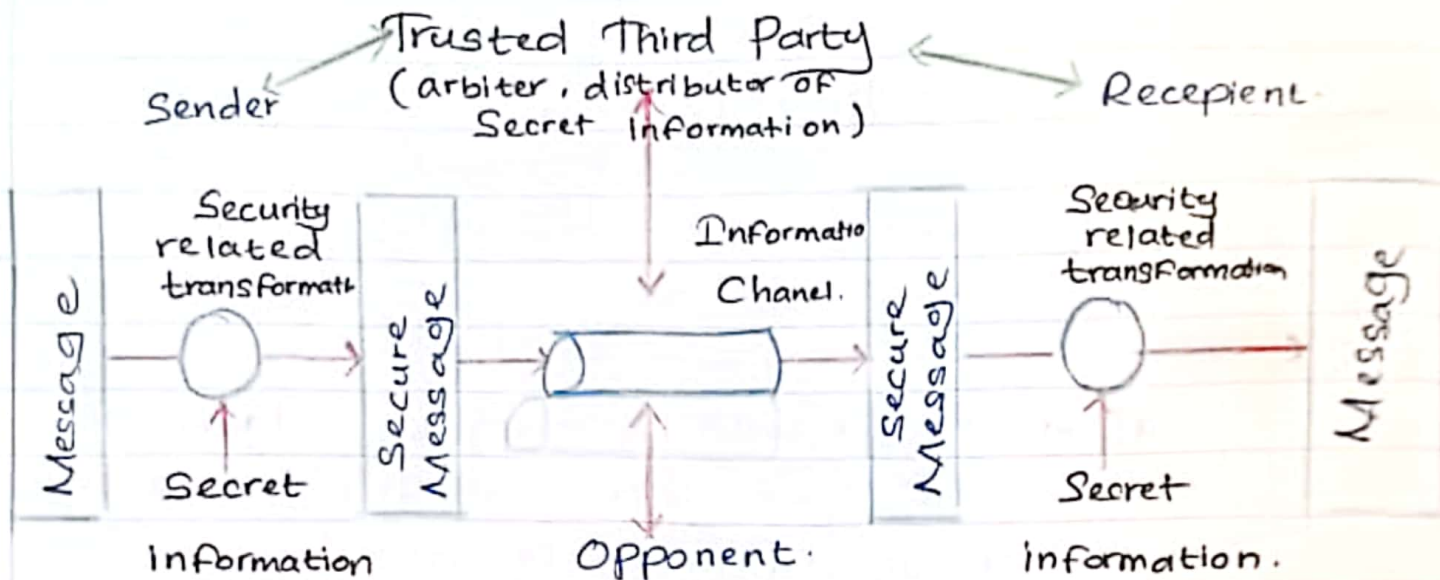
2. RFC 2828

▲ Provide Service Processing or Communication Service provided by System to give a specific Kind of protection to System resources

X.800 provides 5 types of Services.

1. Authentication → Assurance that the communicating entity is the One claimed. (request)
2. Control Access → Prevention of the Unauthorized Use of resources.
3. Data Confidentiality → Protection of data from Unauthorized Person.
4. Data Integrity → Declaration that data received is as sent by and authorize entity.
5. Non repudiation → Protection against denial (rejections). by One of parties in the communication.

Model of network Security.



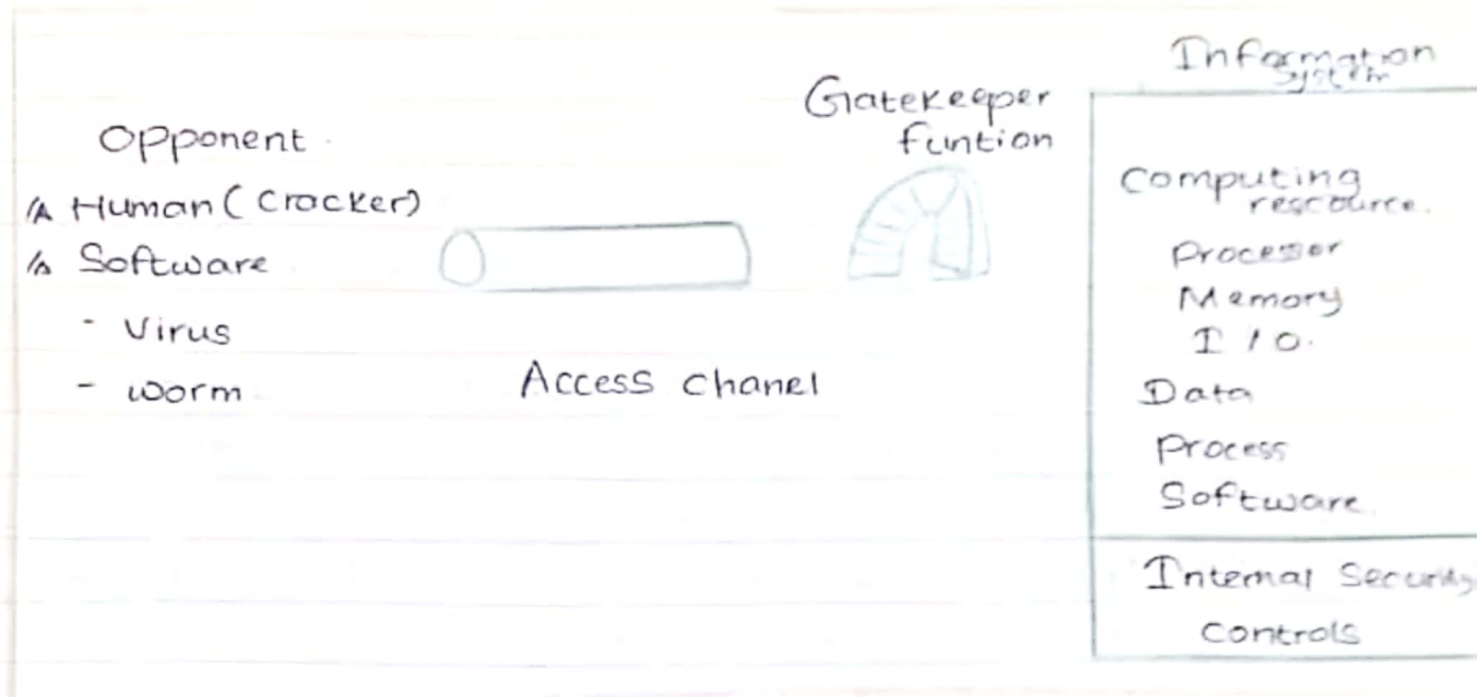
* Design a Suitable algorithm for the Security transformation.

* Generate the Secret information (Key) used by the algorithm.

* Develop method to distribute & share the Secret information. (Cypher text)

* Specify a protocol allow the Principle to use the transformation. & Secret information for a Security Service

Model of Network Access Security



* Using this model required,

1. Select appropriate gate keeper functions (Username passwords, Pin numbers) to identify Users.
2. Implement the Security Controls to ensure only authorize users, access, information or resources.
3. Trusted computer Systems can be used to implement this model.

Types of Security Attack.

1. Passive Security attack.

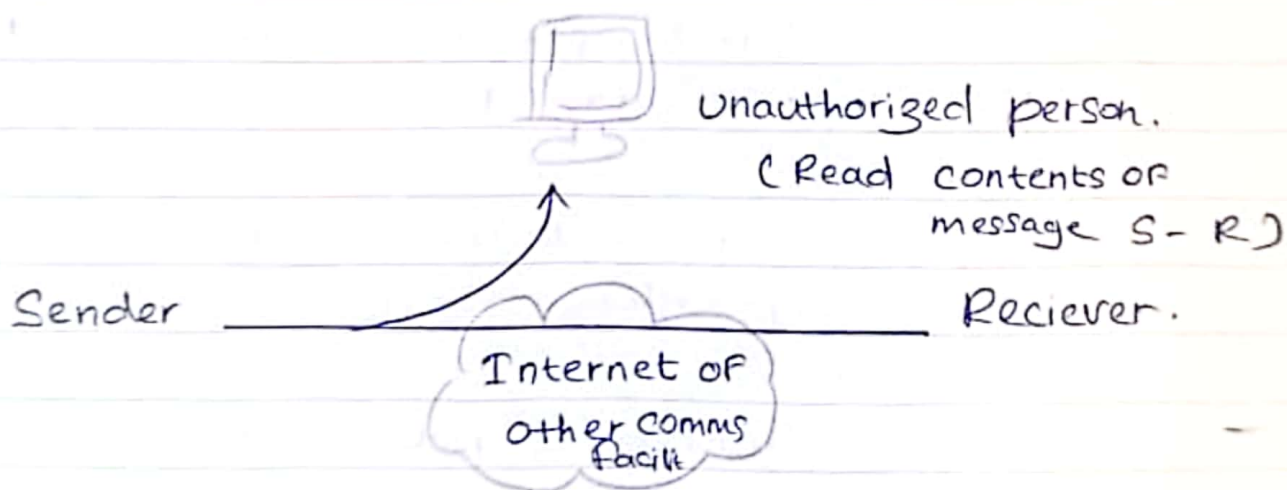
* A passive attack attempt to learn or make use of information from the system but does not affect system resources.

* Passive attackers are in the nature of monitoring of transmissions & read the contents of message.

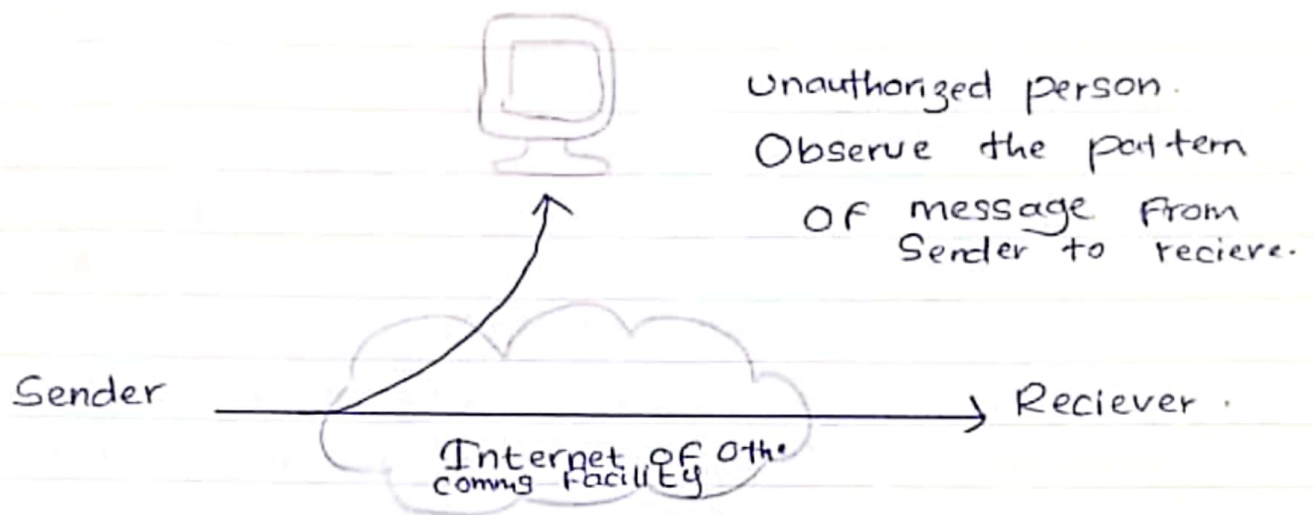
* These attacks are difficult to detect because they do not involve any changes of the data.

* Passive attacks are 2 types.

① Release of message contents.



2. Traffic Analysis

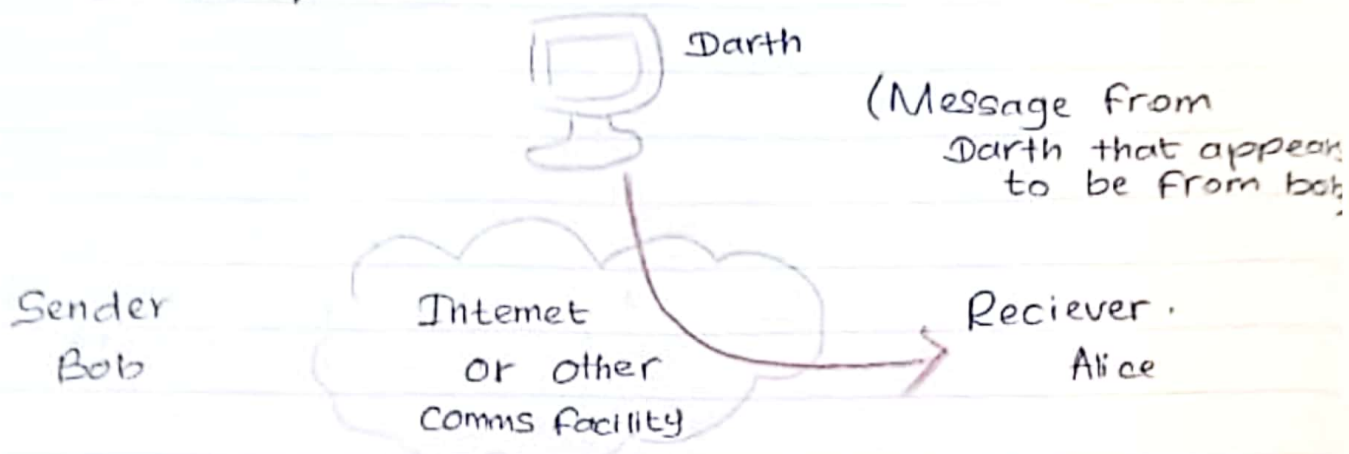


2. Active Security attack.

* Active attacks involves some modification of data stream or the creation of false stream. It can be subdivided into 4 categories.

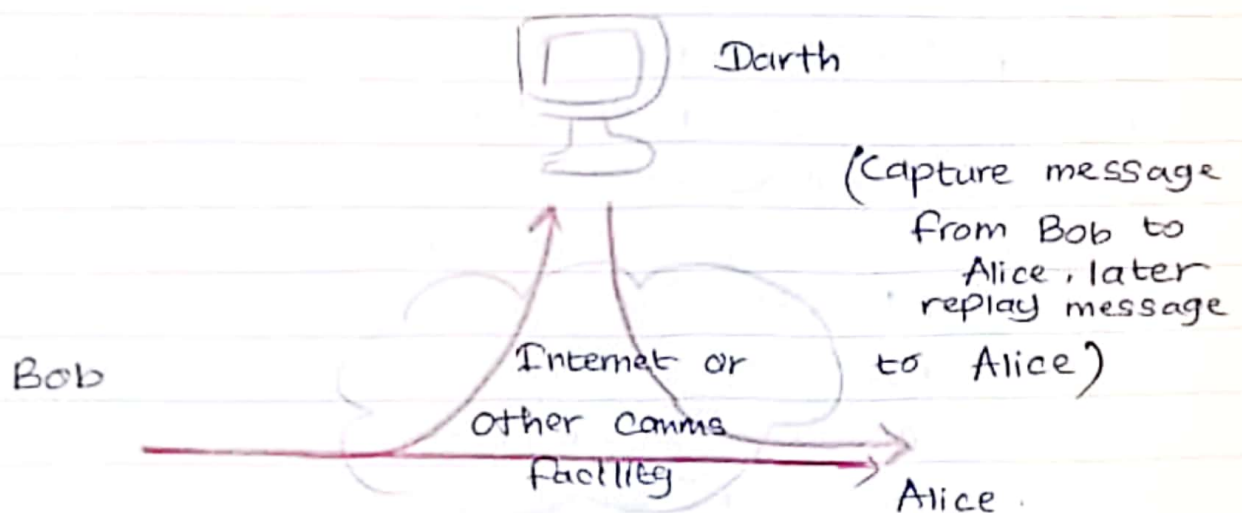
1. Masquerade.
2. Replay.
3. Modification of messages
4. Denial of Service.

① Masquerade.



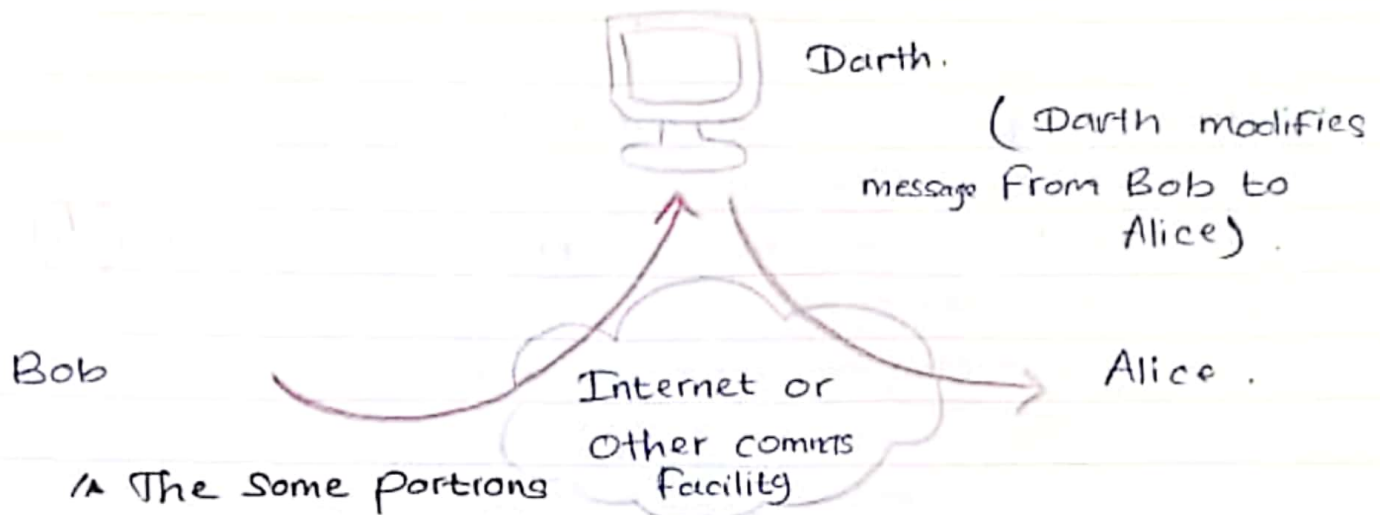
↳ Type of Cyber Security attacks in which an attacker pretend to be someone else in order to gain access to systems or data.

② Replay



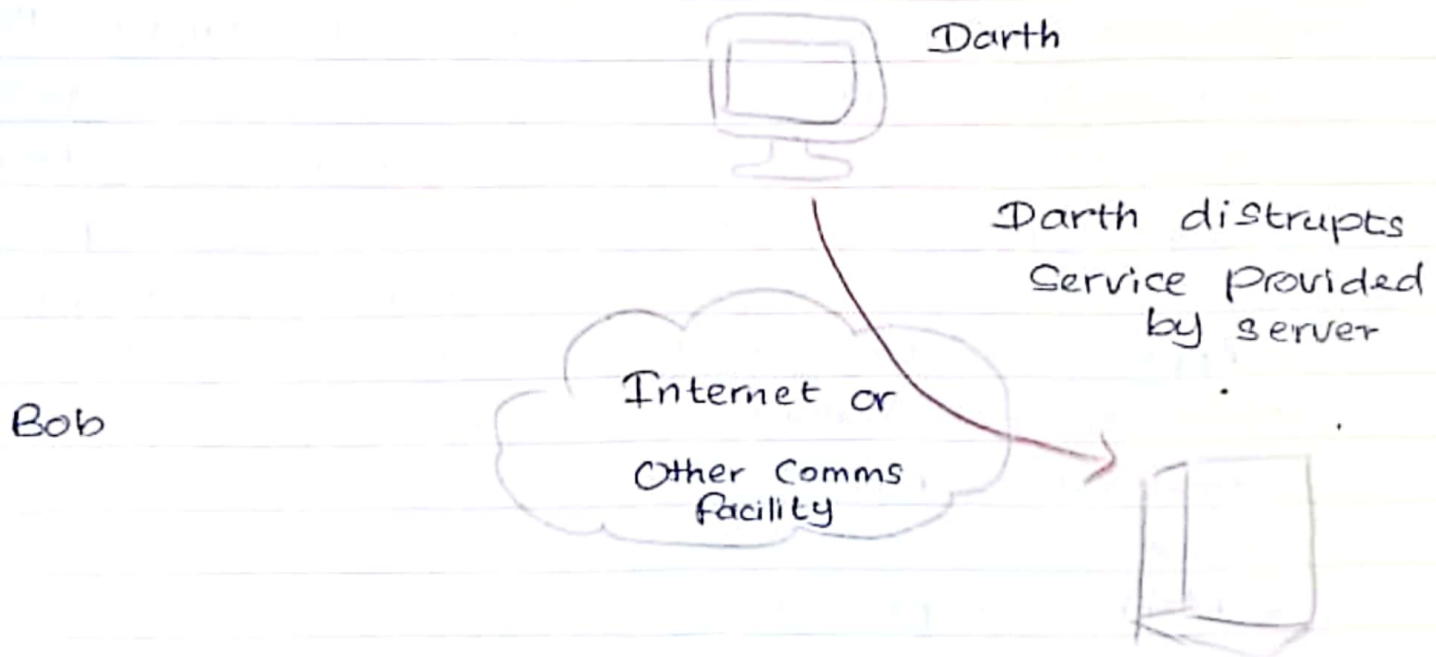
* The basic aim of the attacker is to save the copy of the data originally presented on that specific network & later only use this data for personal use. Corrupted or leak data it is insecure & unsafe for the users.

3. Modification of messages



▲ The some portions of message is changed or that message is delayed or reorder to produce an unauthorized effect. Modification is an attack on the integrity of the original of data

4. Denial of Service. (DOS)



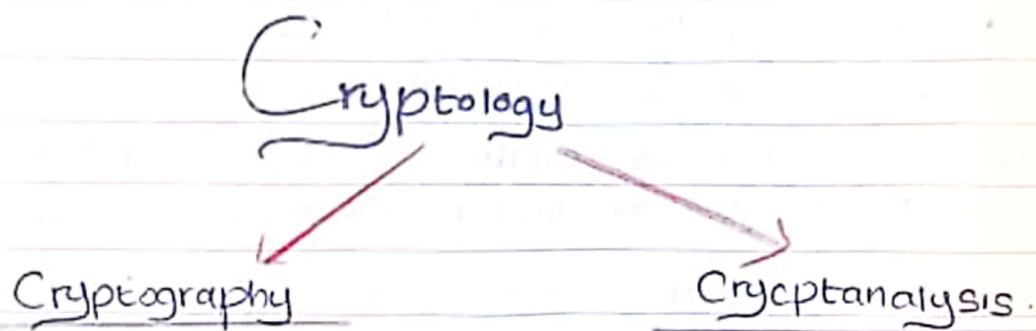
▲ It is a type of cyber security attack. That is designed to make a system or unavailable for authorized users. Sometimes can be make high traffic.

Eg. Flood Attacks.

Flood attacks → Attacker Send large no. of files or request to target system.

How to prevent DoS attack.

- ▲ Using Firewall.
- ▲ Intrusion detection systems to monitor the network traffic.
- ▲ Limiting No. of requests.



* Area of Constructing Cryptographic Systems

▲ The study of encryption principle/methods.

* Apply the data security.

* The Study of Securing Communications from outside Observers

* Method of protecting information & communication through the use of code.

* Code breaking is the area of breaking

Cryptographic Systems.

▲ The Study of principal methods of deciphering cipher text without knowing key.

Security Services of Cryptography.

* Tools & techniques in Cryptography that can be selectively used to provide a set of desired Security services.

- Encryption
- Hash Functions.
- Message Authentication Code.
- Digital Signature

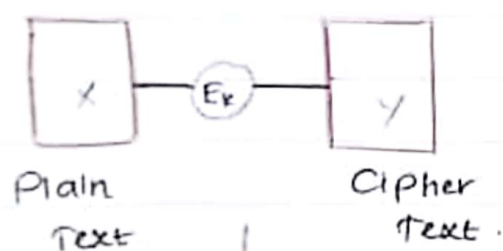
Primitive Services.	Encryption	Hash functions	MAC	Digital Signature
Confidentiality	Yes	No	No	No.
Integrity.	No	Sometimes	Yes	Yes.
Authentication	No	No	Yes	Yes
Non repudiation.	No	No	Sometimes	Yes

$$C = P + K$$

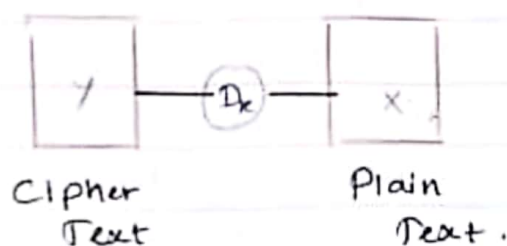
$$P = C - K.$$

Secret Key

$Y = E_k(X)$	Cipher Text	Encryption	Plain Text
$X = D_k(Y)$	Plain Text	Decryption	Cipher Text



Communication media.



Y - Cipher Text.

X - Plain Text.

E - Encryption.

D - Decryption.

Symmetric Encryption.

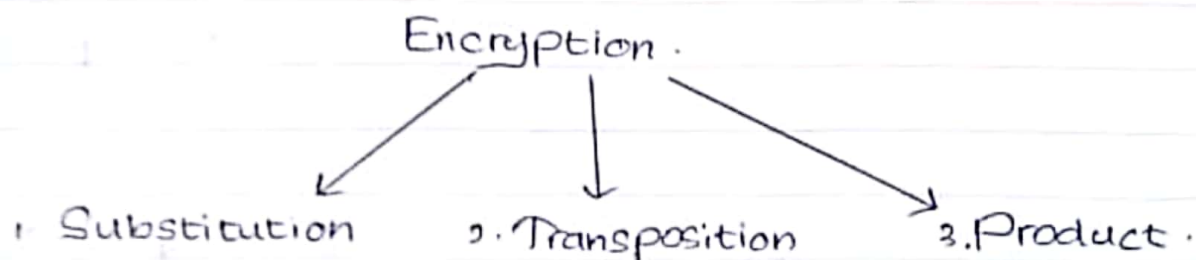
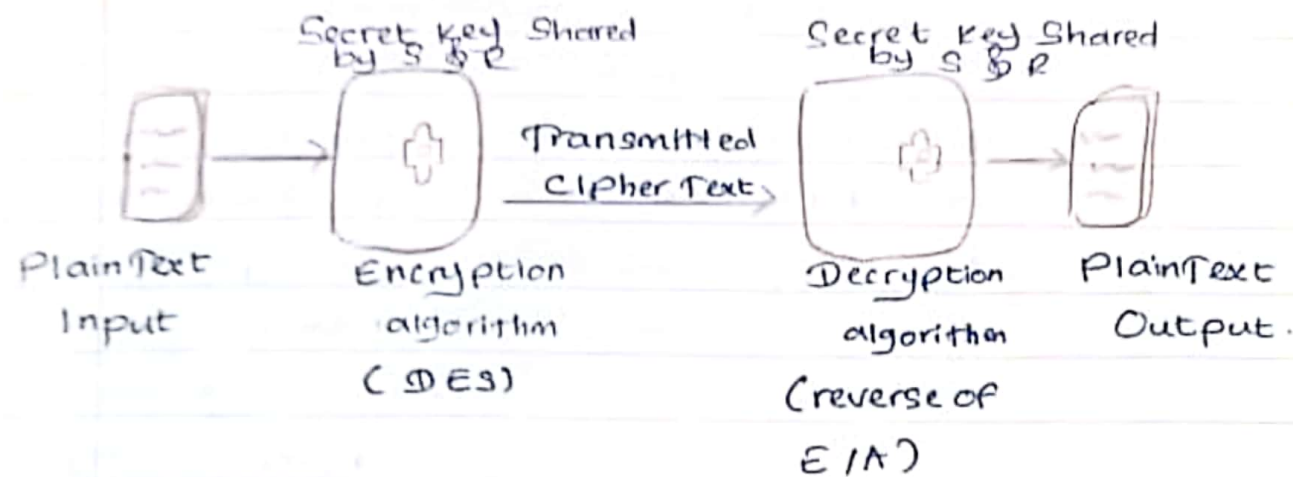
- * Traditional encryption Scheme.
- * Symmetric / Private / Single Key encryption
- * Sender and recipient both use Same Key.

Plain text	- the Original message.
Cipher text	- the Coded message.
Cipher	- Algorithm for transforming P/T → C/T
Key	- Information used in cipher known as only sender/receiver.
Encrypt	- P/T → C/T.
Decrypt	- C/T → P/T.

1. Symmetric Cipher Model

5 Ingredients of Cipher model

- ▲ Plain Text
- ▲ Encryption Algorithm - Perform Substitution On P/A
- ▲ Secret Key - Control exact Substitutions Used in E algorithm
- ▲ Cipher Text
- ▲ Decryption Algorithm - Inverse of encryption algorithm



Symmetric encryption - Single Private

Asymmetric encryption - Two Public

Plain Text is process - Block / Stream

Encryption

1. Substitution

- ▲ Caesar cipher
- ▲ Mono alphabetic
- ▲ Poly alphabetic

2. Transposition

- ▲ Rail fence
- ▲ Row transposition cipher

3. Product

- ▲ 2 Substitution
- ▲ 2 Transposition

Data encryption Scheme in which Units of plaintext are replaced with other Symbols or Group of Symbols

Simple Data encryption Scheme in which Plaintext characters are shifted in some regular pattern to form Ciphertext (Re-order)

Data encryption Scheme in which the Ciphertext produced by encrypting a Plaintext document is subjected to further encryption. By combining 2 or more Simple transposition Ciphers or Substitution Ciphers, a more Secure encryption may result.

1. Substitution

A Plain text replace by letter, No. or Symbols
it involve replacing plain text bit pattern with cipher text bit pattern.

Eg: Ceaser Cipher

1. Ceaser cipher.

* Earliest Substitution Cipher introduce by Julius Ceaser.

* Cipher method is used in ARPA (Advanced Research project Agency) network for military affairs.

* Plain text replace with key.

Eg: $K = 3$

Mathematically give each letter a No.

a	b	c	d	e	f	g	h	i	j	k	l
0	1	2	3	4	5	6	7	8	9	10	11
m	n	o	p	q	r	s	t	u	v	w	x
12	13	14	15	16	17	18	19	20	21	22	23
y	z										
24	25										

Then have Caesar cipher as,

$$C = E(P) = (P + K) \bmod (26)$$

$$P = D(C) = (C - K) \bmod (26)$$

Example

Plain Text	n	e	w		p	a	s	s	w	o	r	d		z	y	a
P	13	4	22		15	0	18	18	22	14	17	3		25	24	0
P + K	16	7	25		18	3	21	21	25	17	20	6		28	27	3
(P + K) mod(26)	16	7	25		18	3	21	21	25	17	20	6		2	1	3
Cipher Text	Q	H	Z		S	D	V	V	Z	R	U	G		C	B	D
Cipher Text	Q	H	Z		S	D	V	V	Z	R	U	G		C	B	D
C	16	7	25		18	3	21	21	25	17	20	6		2	1	3
(C - K) mod(26)	13	4	22		15	0	18	18	22	14	17	3		-1	-2	0
P	13	4	22		15	0	18	18	22	14	17	3		25	24	0
Plain Text	n	e	w		p	a	s	s	w	o	r	d		z	y	a

Plain Text	m	i	s	s	i	o	n		s	u	c	c	e	s	s
P	12	8	18	18	8	14	13		18	20	2	2	4	18	18
P + K	17	13	23	23	13	19	18		23	25	7	7	9	23	23
(P + K) mod(26)	17	13	23	23	13	19	18		23	25	7	7	9	23	23
Cipher Text	R	N	X	X	N	T	S		X	Z	H	H	J	X	X

Crypto Analysis for Ceaser Cipher →

(Brute Force Search).
Crypto Analytic Attack.

▲ Find the plain text from cipher text.

(Crypto Analysis)

▲ You need to recognize the key & decrypt the cipher text into plain text.

GCUA	VQ	DTGCM
EASY	TO	BREAK.

$K = 2$

Mono Alphabetic Cipher.

* Easy to break the Code of Cipher text.

a	b	c	d	e	f	g	h	i	j	k	l	m
D	K	V	Q	F	I	B	J	W	P	E	S	C
n	O	P	q	r	s	t	U	V	w	x	y	z.
x	H	T	M	Y	A	U	O	L	R	G	Z	N

Eg: E A S Y T O B R E A K.
F D A Z U H K N Y P D E.

2. Transposition Cipher.

* Hide the message by re arranging the letter Order.

* Without changing the actual record use can recognise same frequency as the Original text.

Eg: Rail fence cipher.

Row transposition cipher.

m e e t m e a f t e r t h e t o g a p a r t y

meet me after the togaparty.

Eg: Information Security System.

I n f o r m a t i o n s e c u r i t y s y s t e m

IFraineu1gtytmnomtoscrtsse.

Row Transposition Cipher.

* More Complex cipher Scheme used the Key & write the plaint text row by row according to the given key read column by column with the ascending Order of the key and make the cipher text.

Eg drop is change.

2	4	1	3
d	r	o	p
i	s	c	h
a	n	g	e

Ciper Text = OCGDIAPHERSM.

Password change to chatcat.

2	3	4	1	5
p	a	s	s	w
o	r	d	c	h
a	n	g	e	t
o	c	h	a	t
c	a	t	y	2

Ciper Text = SCEAYPOAOCARNCASDGHHTWHT
TT2.

3. Product Cipher.

* Ciphers using Substitution or Transposition are not secure because of language characteristics.

* Hence consider using several ciphers

in succession to make harder but,

← 2 Substitutions make a more complex Substitution

← 2 Transpositions make a more complex Transposition

← but a Substitution followed by a transposition makes a new much harder cipher.

* This a bridge from classical to modern crypt.

Message :

01. Drop point? is closed until tomorrow.

- Caesar cipher ($K=4$)
- Mono Alphabetic Ciphers.

02. 2 Transposition ciphers

- Row transposition (key = 4132) & rail fence.

03. Substitution & Transportation.

- Caesar cipher (key = 5) & rail fence.

04. Transposition & Substitution.

- Transposition (key = 54132) & Caesar Ciphers

Question - 01

Plaint/r	D	R	O	P	P	O	I	N	T	I	S	C	L	O	S	E	D
P	3	17	14	15	15	14	8	13	19	8	18	2	11	14	18	4	3
D + K	7	21	18	19	19	18	12	17	23	12	22	6	15	18	22	8	7
D + K(mod)	7	21	18	19	19	18	12	17	23	12	22	6	15	18	22	8	7
Cipher/r	H	V	S	T	T	S	M	R	X	M	W	G	P	S	W	I	H

→ Caesar cipher.

HVSTTSMRXMWGP SWIH

JLAUVACYGCRBTARSTJ → MonoAlphabetic.

Question 02.

4 1 3 2
d r o p
p o i n c
t l s c
l o s e
d l x y 2.

Transposition cipher = ROIXPNCE201SSYDPTID

Rail fence =
R I X N E O S Y P I
O O P C 2 I S D T
= RIXNEOSYP1OOPC21SDTD

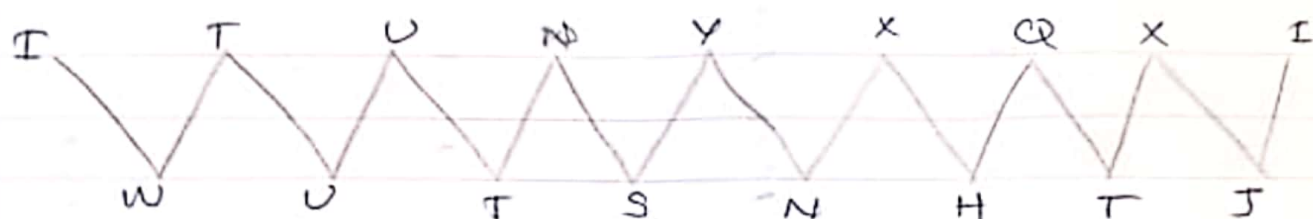
Question 03.

Date

Page

Plaintext	D	R	O	P	P	O	I	N	T	I	S	C	L	O	S	E	D
P	3	17	14	15	15	14	18	13	19	8	18	2	11	14	18	4	3
P+k	8	22	19	20	20	19	13	18	24	13	23	7	16	19	23	9	8
Ciphertext	I	W	T	U	U	T	N	S	Y	N	X	H	Q	T	X	J	I

I W T U U T X S Y N X H Q T X J I → ceaser cipher.



I T U N X X Q X I W U T S N H T J → Rail Fence.

Question 04

5 4 1 3 2.

d r o p p

o i n t i

s c l o s

e d x y 2.

Row transposition = ONLXPISZPTOYRICDDOSE.

Ciphertext	O	N	L	X	P	I	S	Z	P	T	O	Y	R	I	C	D	D	O	S	E
C	14	13	11	23	15	8	18	25	15	19	14	24	7	8	2	3	3	14	18	4
(C-K)	8	7	2	17	9	6	12	19	9	13	8	18	1	2	4	3	-3	8	12	-2
Plaintext	I	H	F	R	J	G	M	T	J	N	I	S	b	C	V	N	N	I	M	X

Plain/T	O	N	L	X	P	T	S	Z	P	T	O	Y	R	I	C	D	D	O
P	14	13	11	23	15	8	8	25	15	19	14	24	17	8	2	3	3	14
(P+K)	20	19	17	29	21	14	24	31	21	25	20	30	23	14	8	9	9	2
Cipher/T	U	T	R	D	V	O	Y	F	V	Z	U	E	X	O	I	J	J	U

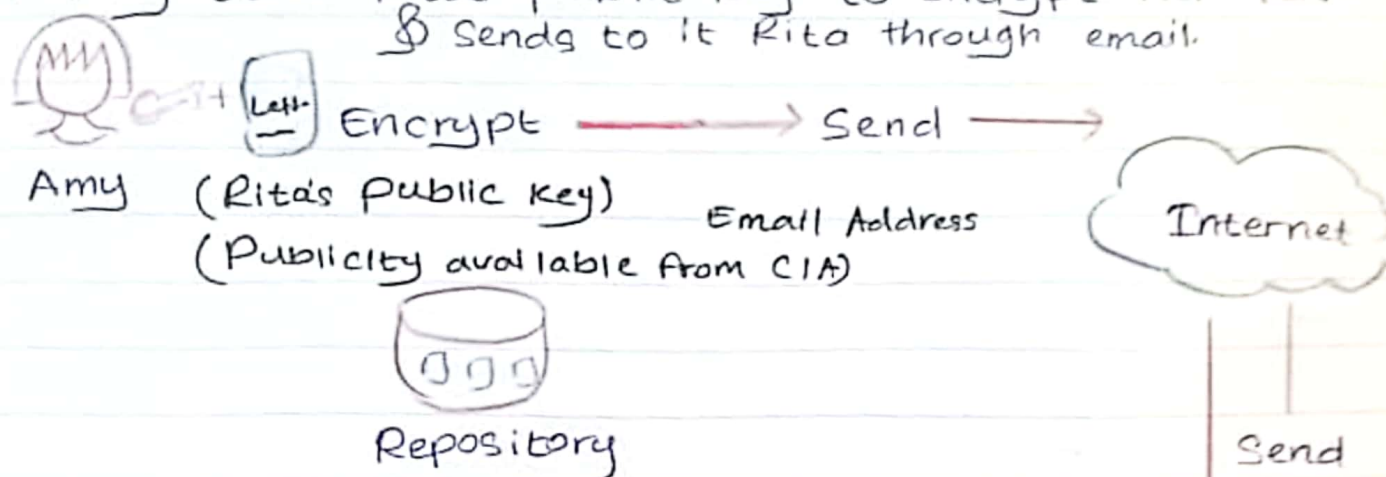
Plain/T	S	E
P	18	4
(P+K)	24	10
(cipher/T)	Y	K

Ceaser Cipher = UTRDV OYFVZ UEX OIJTUYK

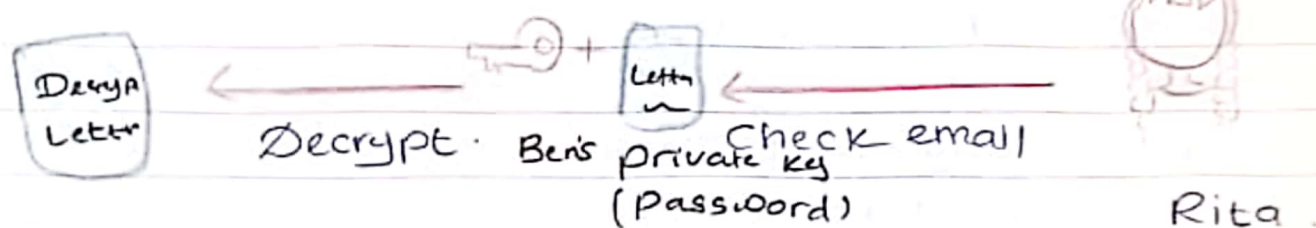
2. Asymmetric Cipher Model.

- ▲ Probably most Significant advance in 3000 Year history of Cryptography
- ▲ Uses 2 Keys Public & Private Key.
- ▲ Asymmetric Since parties are not equal.
- ▲ Uses clever application of number theoretic Concepts to functions.
- ▲ Complement rather than private key crypto.

1. Amy uses Rita's public key to encrypt her letter & sends it to Rita through email.



2. Rita upon receiving the email, decrypts Amy's email with his own private key



Why Asymmetric Encryption

- * Developed to address 2 key issues.

- Key distribution.

How to have secure communications in general without having to trust a KDC with ur key.

- Digital Signature. (Receiver)

How to verify a message comes

intact from the claimed sender. (Authority who sends)

- * Public invention due to Whitfield Diffie & Martin Hellman at Stanford Uni in 1976.

- Known earlier in classified communications

Public Key Cryptography.

* Public key / two keys / asymmetric cryptography involves the use of 2 keys.



- Anybody can be used to encrypt messages, and verify messages (Signatures)
- The recipient used to decrypt messages, and Sign(create) Signatures.

* Is asymmetric bcz, those who encrypt messages or verify Signatures cannot decrypt messages or Create Signatures.

Characteristics.

check the encrypted key that is tally to private keys.

* Public Key rely on 2 keys.

- It is computationally infeasible to find decryption ^(private) key knowing only algorithm and encryption key.

- It is computationally easy to encrypt / Decrypt messages when the relevant (encryption / Decryption) key.

- Either of the 2 related keys can be used for encryption, with the other use for decryption.

Way in which Plain
text process

* Block

* Stream

Public Key Applications

Encryption /
Decryption
(Receiver)

Digital Signature
(Sender)

Key exchange
(Receiver)

↳ Provide Secrecy

↳ Provide authentication

↳ Session key

↳ Some Algorithms are Suitable for all uses, others are specific to one.

Security Threats.

Disclosure

Traffic analysis.

Modification.

Message Authentication Key (Code)

- * Small
- * Fixed size.
- * Not reversible.
- * Both message and key.

Message Authentication

Message authentication is concerned with,

1. Protecting the integrity of a message.
2. Validating identity of Originator.
3. Non repudiation of Origin.

* Pre - authentication functions

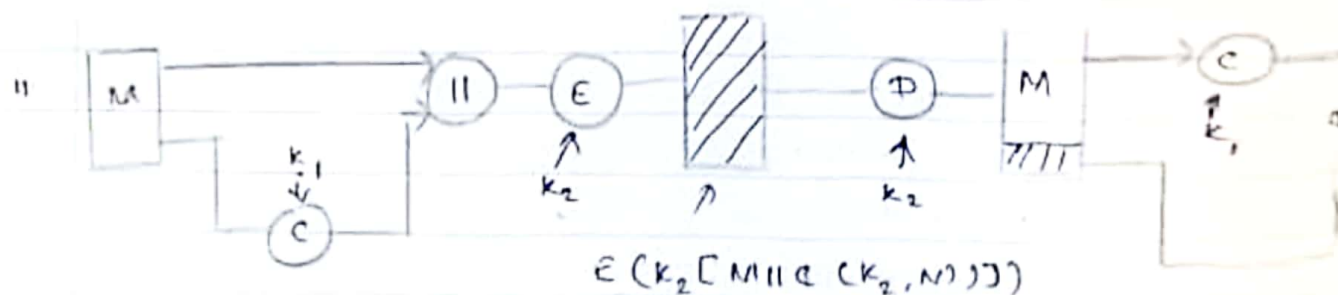
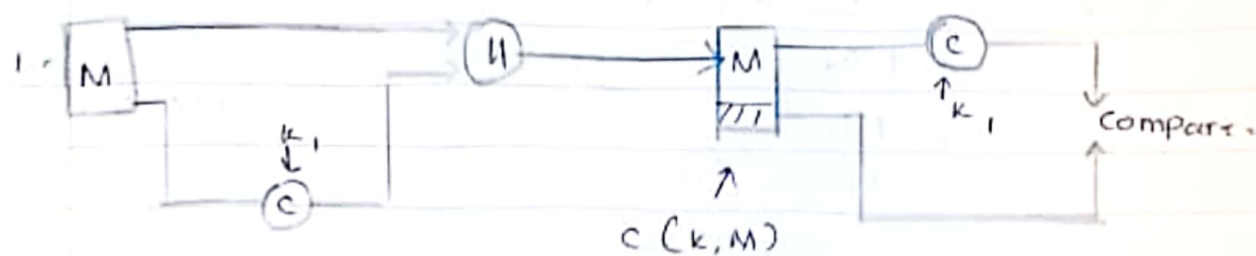
1. Message encryption.
2. Message authentication Code. (MAC)
3. Hash function.

Security requirements.

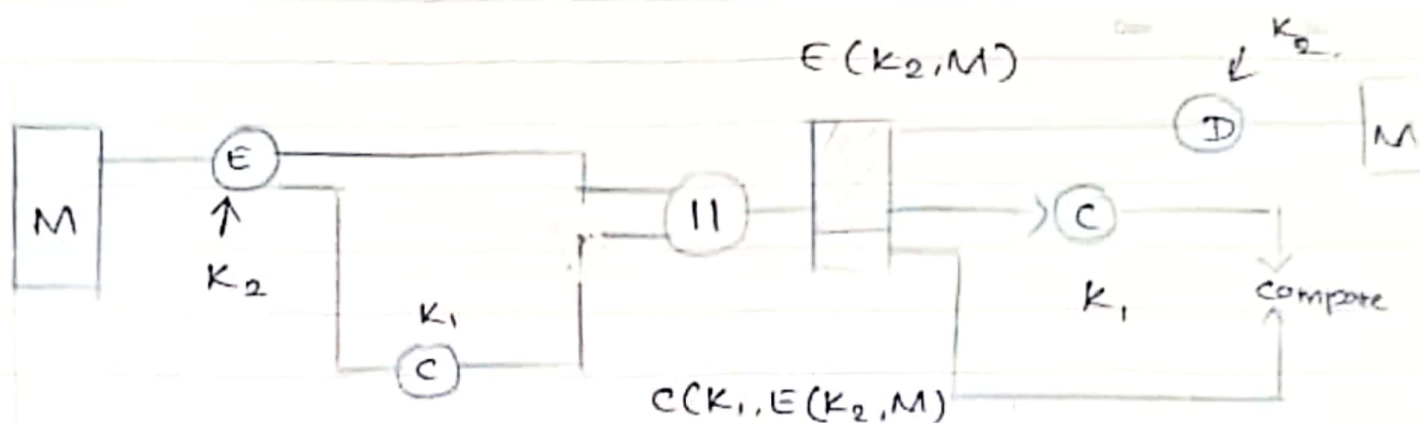
1. Disclosure.
2. Traffic Analysis.
3. Masquerade.
4. Content modification.
5. Sequence modification.
6. Timing modification.
7. Source repudiation.
8. Destination repudiation.

Message Authentication Code, (MAC)

- * Message and Some Key Create Small Fixed Sized block. (MAC)
- * MAC not be reversible.
- * Appended to a Signature.
- * Receiver perform Same Computation on message & key, make MAC; check ^{Sender} receiver MAC & receiver MAC and confirm the authentication.
- * MAC provide assurance that message is unaltered & comes from Sender.
- * MAC also type of cryptography used authentication technique. It involve Secret Key & message to generate a small fixed sized block of data.



iii



- * MAC Provide authentications
- * Use encryption for Security.
- * Use one or more keys for generate the MAC and encryption.
- * Can Compute MAC either before or after encryption.
- * Why used a MAC?

- Sometimes only authentication is needed.
- Sometimes need authentication to Continue longer than the encryption.

* MAC is not a digital Signature.

MAC Properties

- * A MAC is a Cryptographic checksum.
 $MAK = C_K(M)$
- * Condenses a variable length message M .
- * Using a Secret key K .
- * To a Fixed sized authenticator.

* Its a many to one function.

- Potentially many messages have same MAC.
- Difficult.

Requirements for MAC.

1. Knowing a message & MAC. IS infeasible to find another messages with same MAC.
2. MAC should be uniformly distributed.
3. Mac should depend equally on all bits of message.

Authentication.

- ▲ Granting the approval.
- ▲ Allow or restrict access to data or devices.

Access control Terminology.

1. Identification.
2. Authentication.
3. Authorization.

Basic steps in access control.

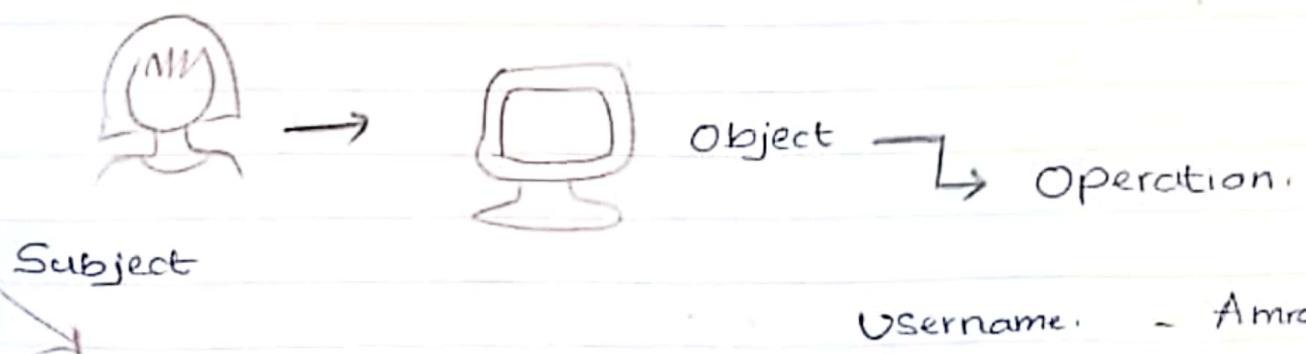
Identification	User enters User name.
Authentication	User provides a password
Authorization	User authorized to login.
Access	User allowed to access only specific data.

Access Control Terminology

1. Object → Resources
2. Subject → LiveWare.
3. Operation → Both changes. (Resources + LiveWare).

Subject

- owner → Director. Responsible for information.
- Custodian. → Lectures. day to day action ← assigne
- End User. → Students who access.



Username. - Amra.
Password - ****
Authorization - HeyHi
Access - OpenAmu

Access Control Models.

Four major Model.

1. Mandatory Access Control.
2. Discretionary Access Control (Based on owner) (Least restrictive).
3. Role based access Control
4. Rule based access Control.

	Security Model	Lattice Model	Bell - Lapadula Model.
1.	Organization	• Security levels Organized in a lattice structure	• Based on a lattice model with a specific access control rules
2.	Partial Ordering	• Represents a partial ordering of security levels. (Not all level needs to be directly compared)	• Defines relationship between security levels.
3.	Access Control	• Determines access by comparing Subjects & Object level.	• Controls access based on security clearances.
4.	Information flow	• Allows upward information flow, restricts downward flow.	• Emphasizes preventing Unauthorized information disclosure.
5.	Security levels.	• Associated with a set of Permissions.	• Subjects and Objects have security classification
6.	Simple Security Property.	• No read up Subjects can't read high level data.	• Prevents reading data at a higher security level.

7.	Star Property	- No write down - Subjects can't write lower level data	Prevents writing data to a lower security level.
8.	Access Control rules	No specific rules, depends on lattice comparisons	Enforces rules to prevent information leakage.

Noted → * Can apply to the System.

Information

Security (Database)

3. Role Based Access Control. → Based on the User role.
4. Rule Based Access Control. → Based on the role of the responsibility rule.

Example for best practices in Access Control.

- ▲ Job rotation.
- ▲ Separation of duties.
- ▲ Least privileges.
- ▲ Implicit deny.
- ▲ Mandatory Vacations.